A finite grouped-wait assay for direct intervention decisions

Astra construction: requested operating-rate interface

5 November 2026

Abstract

We construct the arithmetic mean of unknown heterogeneous local operating escape rates without recovering their full table. Four first-exit waits at one randomly chosen requested context give a classical inverse-gamma rate statistic with finite variance. Explicit clipping, censoring, first-event timing, finite arithmetic and conditional service errors yield a bounded physical trial whose second moment remains independent of the requested accuracy. A conditional median-of-means proof gives quadratic precision cost even for repeated cells, without assuming independent hardware errors. The complete dwell bill includes discarded censored trials. A two-menu rare-target family gives a matching quadratic total-exposure lower bound even with full local trajectories, adaptive calibrated fields and free parameter-independent resets.

1 A finite direct assay for the mean boundary escape rate

The required decision statistic is an arithmetic average of actual local operating rates. For a menu setting u and target configuration b, let

$$\lambda_i^u = Q^u(b, b^{(i)}), \qquad \mu_u = \frac{1}{N}\sum_{i=1}^{N} \lambda_i^u, \qquad \Lambda_u = N\mu_u.$$

Suppose a common certified envelope supplies

$$0 < \lambda_- \le \lambda_i^u \le \lambda_+ < \infty, \qquad \kappa = \lambda_+/\lambda_-. \tag{1}$$

For the preserved binary Gibbs kinetics with $b_i = 1$, $\lambda_i^u = r_i^u(b_{\partial i})[1 - q_i^u(b_{\partial i})]$. Thus $r_i^u \ge r_-$ and $q_i^u \le \bar{q} < 1$ give the usable fast-direction floor $\lambda_- = r_-(1-\bar{q})$, and $r_i^u \le r_+$ gives $\lambda_+ = r_+$. These are bounds on the requested operating environments, not rates transported from balancing fields. Random-context averaging does not establish a uniform bound on unvisited contexts. A supplied engineering bound or an independently acquired uniform certificate retains its own full acquisition cost.

For each trial choose a coordinate uniformly, install its declared neighborhood in the target context on a faithful local copy, and make four successive pin–release first-exit observations from the same requested starting bit. In the ideal Markov model the four waits at coordinate i are independent $\mathrm{Exp}(\lambda_i^u)$ conditional on the requested coordinate. Repeated physical cells need only the uniform conditional implementation guarantees below. No independence between imperfect trials or cells is assumed.

The context setting, four starting-state preparations, settling, operating-field reproduction, detector operation and clock service are all paid. In a bounded-neighborhood model this sets only the declared local neighborhood; it does not prepare the entire global target state. A local pin is not a free natural visit to that state. Structural closure and faithful local replication remain premises.

1.1 The classical inverse-gamma statistic and finite modifications

Let $T_1, \dots, T_4$ have rate λ and set $S = \sum_j T_j$. The classical unbiased rate statistic is $X = 3/S$. Direct integration of the gamma density gives

$$f_S(s) = \frac{\lambda^4 s^3 e^{-\lambda s}}{6}, \qquad \mathbb{E}X = \lambda, \qquad \mathbb{E}X^2 = \frac{3}{2}\lambda^2. \tag{2}$$

This identity is standard exponential-family estimation, not an invented estimator; it is derived here to expose the exact variance and finite device requirements. In particular, pooling heterogeneous waits and inverting their overall mean would estimate a harmonic-rate quantity, not the required arithmetic mean. The coordinate is held fixed within each group of four, then the resulting rate statistics are averaged.

Fix $0 < \eta \le 1/4$. Choose a rational a with

$$\frac{\eta^{1/3}}{2\lambda_+} < a \le \frac{\eta^{1/3}}{\lambda_+}, \qquad B = 3/a, \qquad D = \frac{\lceil \log_2(192/\eta) \rceil}{\lambda_-}. \tag{3}$$

A dyadic $a\lambda_+$ satisfying these inequalities is obtained by integer comparisons of cubes. Let

$$X_a = \min(3/S, B) = 3/\max(S, a), \qquad Z = X_a \mathbf{1}_{\{\max_j T_j \le D\}}.$$

Run the waits sequentially, stopping and returning zero as soon as one wait is censored. The ideal output lies in $[0, B]$. The two lost-mean bounds are

$$0 \le \lambda - \mathbb{E}X_a \le \frac{\lambda^4 a^3}{24} \le \frac{\eta\lambda}{24}, \tag{4}$$

$$0 \le \mathbb{E}X_a - \mathbb{E}Z \le \frac{12 e^{-\lambda D}}{D} \le \frac{\eta\lambda_-}{16}. \tag{5}$$

For (4), integrate $(3/s - 3/a) f_S(s)$ on $[0, a]$ and bound $e^{-\lambda s}$ by one. For (5), on any $T_j > D$ one has $X_a \le 3/D$; take a union bound over four waits. Also $D\lambda_- \ge 1$ and $e^{-\lambda_- D} \le 2^{-\lambda_- D} \le \eta/192$. The clipped, censored second moment is at most $3\lambda^2/2$.

1.2 Actual deadlines, first-event timestamps and finite arithmetic

An actual timer must supply the following service, including start, release, detector latency and deadline response in the error budget: outside a declared bad-service event, an uncensored first-exit timestamp differs from the true wait by at most Δ, and each actual censor deadline differs from D by at most Δ. Take

$$0 < \Delta \le \frac{\eta\lambda_- a^2}{1024}, \qquad 0 < \omega \le \frac{\eta\lambda_-}{128}, \qquad 0 \le \beta \le \frac{\eta\lambda_-}{128B}. \tag{6}$$

ω bounds final numerical rounding of the rate statistic. The total conditional TV or bad-service probability allowance β per trial covers finite coordinate selection, context and pin failures, missed exits, nonideal local conditional laws and other declared stochastic readout errors. It is a guarantee conditional on the entire past, on the common calibration good event. Deterministic timer errors are accounted for separately and are not represented as a false TV approximation of a continuous timestamp by a grid.

The reported statistic is $Y = 3/\max(a, \sum_j \widehat{T}_j)$ if all four observations are accepted, and zero on a censor or flagged failure, rounded within ω and clipped to $[0, B]$. Round each timestamp with its physical errors included in Δ. Exact rational division is also allowed. Every operation has a finite cap, including failed reads and state preparations.

The censor decision can disagree with its ideal value only when at least one true wait lies in $[D - \Delta, D + \Delta]$, or on the bad-service event. The ideal censor-boundary band has probability

at most $8\lambda_+\Delta$. Away from that band, the reciprocal clipping map is $3/a^2$-Lipschitz, so its time-induced error is at most $12\Delta/a^2$. Consequently

$$|\mathbb{E}(Y \mid \mathcal{F}) - \mathbb{E}Z| \le \frac{12\Delta}{a^2} + 8B\lambda_+\Delta + B\beta + \omega. \tag{7}$$

Here $\mathcal{F}$ contains every previous trial and acquisition outcome; the ideal expectation includes a fresh uniform coordinate. The band term is essential: a Lipschitz bound on a recorded sum alone would not account for censor decisions changed by timer error.

Lemma 1 (Uniform conditional moments of the physical trial)**.** *On the simultaneous calibration good event, the construction satisfies*

$$|\mathbb{E}(Y \mid \mathcal{F}) - \mu_u| \le \frac{\eta\mu_u}{6}, \qquad \mathbb{E}(Y^2 \mid \mathcal{F}) \le 4\lambda_+\mu_u \le 4\lambda_+^2. \tag{8}$$

Proof. Average (4) over the uniform coordinate. The clipping loss is at most $\eta\mu_u/24$. The censor loss is at most $\eta\lambda_-/16$. Since $\lambda_+ a \le 1$, the remaining errors in (7) sum to at most

$$\eta\lambda_- \left(\frac{12+24}{1024} + \frac{1}{128} + \frac{1}{128}\right) = \frac{13\eta\lambda_-}{256}.$$

Thus the total is at most $\eta\mu_u(1/24 + 1/16 + 13/256) = 119\eta\mu_u/768 < \eta\mu_u/6$.

For the second moment couple to the ideal trial. Away from the censor band and bad-service event, the output differs from Z by at most $e = 12\Delta/a^2 + \omega \le 5\eta\lambda_-/256$. Hence

$$\mathbb{E}(Y^2 \mid \mathcal{F}) \le 2\mathbb{E}Z^2 + 2e^2 + B^2(8\lambda_+\Delta + \beta).$$

The first term, averaged over the requested coordinate, is at most $3N^{-1}\sum_i \lambda_i^2 \le 3\lambda_+\mu_u$. The band contribution is at most $72\eta\lambda_-\lambda_+/1024$. The lower bound on a gives $B < 6\lambda_+\eta^{-1/3}$, so $B^2\beta \le 6\eta^{2/3}\lambda_-\lambda_+/128$. Together with $\eta \le 1/4$ these terms and $2e^2$ are strictly below $\lambda_+\lambda_- \le \lambda_+\mu_u$. This proves the bound without replacing the variance by the much larger worst-case range B^2. □

The timer tolerance scales as $\Delta = \Theta(\eta^{5/3}\lambda_-/\lambda_+^2)$ for a comparable choice in (6). Its response and calibration price remain physical costs. The number of timestamp and arithmetic bits is $O(\log(1/\eta) + \log\kappa + \log\log(1/\eta))$ beyond the rate and hardware scale representation. This bit count does not establish that a device can service that resolution cheaply.

1.3 A finite confidence bound with dependent repeated cells

Theorem 2 (Direct acquired arithmetic-rate certificate)**.** *For each of K declared menu settings, divide the trials into q blocks of size m, where*

$$m = \lceil 256\kappa\eta^{-2} \rceil, \qquad q = \textit{the smallest positive odd integer with } 2^{-q} \le \delta/K. \tag{9}$$

Return the median $\widehat{\mu}_u$ of the q block averages. Conditional on the common calibration good event, with probability at least $1-\delta$, simultaneously for all menu settings,

$$|\widehat{\mu}_u - \mu_u| \le \eta\mu_u. \tag{10}$$

In fact the proof gives the stronger constant $2\eta/3$ on that event. The number of trials per menu is $O(\kappa\eta^{-2}\log(K/\delta))$, independent of N except for coordinate selection, context access and their actual service costs. No kinetic rate table, rare stationary count or global trajectory is acquired by this assay.

Proof. For one block, write $Y_j = m_j + D_j$ with $m_j = \mathbb{E}(Y_j \mid \mathcal{F}_{j-1})$ and (D_j) martingale differences. The conditional second-moment bound gives $\mathbb{E}[(\sum_j D_j)^2 \mid \mathcal{F}_{\text{start}}] \le 4m\lambda_+\mu_u$. Conditional Chebyshev therefore implies

$$\mathbb{P}\left(\left|m^{-1}\sum_j D_j\right| > \frac{\eta\mu_u}{2} \,\middle|\, \mathcal{F}_{\text{start}}\right) \le \frac{16\lambda_+}{m\eta^2\mu_u} \le \frac{16\kappa}{m\eta^2} \le \frac{1}{16}.$$

Every m_j differs from μ_u by at most $\eta\mu_u/6$. Thus each block average is within $2\eta\mu_u/3$ except on an event whose conditional probability, given all preceding blocks, is at most $1/16$. For any specified set of s bad blocks, repeated conditioning bounds their joint probability by 16^{-s}. A bad median needs at least $\lceil q/2 \rceil$ bad blocks, so its probability is at most $2^q 16^{-\lceil q/2 \rceil} \le 2^{-q}$. No independence of blocks is used. Union-bound over the K menus; adaptive scheduling is allowed provided the menus and trial guarantees are the declared ones. □

Theorem 3 (A smaller direct-mean schedule)**.** *The same implemented trials can instead be averaged directly. With*

$$n = \left\lceil 40\kappa\eta^{-2}\log(2K/\delta) \right\rceil \tag{11}$$

trials per menu, their ordinary average satisfies (10) *simultaneously with probability at least* $1-\delta$*, conditional on calibration. The bound again holds for arbitrarily dependent repeated cells satisfying the conditional service guarantees. No empirical or independence approximation is used.*

Proof. For the martingale differences $D_j = Y_j - \mathbb{E}(Y_j \mid \mathcal{F}_{j-1})$, $|D_j| \le B$ and the conditional variance is at most $v = 4\lambda_+\mu_u$. For $|s|B < 3$, expanding the exponential and using $k! \ge 2 \cdot 3^{k-2}$ for $k \ge 2$ gives

$$\mathbb{E}(e^{sD_j} \mid \mathcal{F}_{j-1}) \le \exp\left(\frac{s^2 v}{2(1-|s|B/3)}\right).$$

Iterate this conditional bound and apply Chernoff on each side. The resulting standard martingale Bernstein inequality is

$$\mathbb{P}\left(\left|n^{-1}\sum_j D_j\right| > t\right) \le 2\exp\left[-\frac{nt^2}{2(v+Bt/3)}\right].$$

With $t = \eta\mu_u/2$ and $B\eta < 6\lambda_+\eta^{2/3} \le 6\lambda_+$, the exponent is at least $n\eta^2\mu_u/(40\lambda_+) \ge n\eta^2/(40\kappa)$. The conditional-mean bias adds at most $\eta\mu_u/6$, giving total error at most $2\eta\mu_u/3$. Union-bound over the menus. □

If common apparatus or structural calibration fails with probability δ_{cal}, the unconditional confidence is at least $1 - \delta_{\text{cal}} - \delta$. This is distinct from a per-trial conditional error; reusing a persistently miscalibrated device does not generate new independent confidence. A common drift or hidden cell memory not covered by (8) invalidates the stated martingale guarantee.

The reported interval for μ_u is $[\widehat{\mu}_u/(1+\eta), \widehat{\mu}_u/(1-\eta)]$, with outward rounding; multiply by N for Λ_u. Choosing the smallest $\widehat{\mu}_u$ yields

$$\mu_{\widehat{u}} \le \frac{1+\eta}{1-\eta}\min_u \mu_u.$$

If the resulting intervals are disjoint, their ordering is certified. The separate mathematical bridge from boundary escape to the intended global first-event objective must still be proved at its stated scope; the local assay alone does not equate stationary flux with lifetime.

The direct-mean implementation is streaming: it stores four current timestamps, a sum and count, a coordinate index, and the K final intervals. The median variant additionally stores its q

block means. The arithmetic bit lengths are polynomial in rate and hardware bit length, $\log N$, $\log n$, and requested precision. Known coordinate labels and neighborhood access, apparatus cells, context memory and addressing latency remain priced resources; no free global wiring or new physical copies are inferred from this small statistical memory.

1.4 Full dwell and reset bill, including discarded trials

Let $s = e^{-\lambda D}$ and $q_\lambda = 1 - s$ for a requested rate. The ideal sequentially censored trial has expected active dwell

$$\mathbb{E}W_\lambda = \frac{q_\lambda}{\lambda} \sum_{j=0}^{3} q_\lambda^j \le \frac{4}{\lambda}, \qquad W_\lambda \le 4D. \tag{12}$$

The expected number of pin–release observations is $\sum_{j=0}^{3} q_\lambda^j \le 4$. The probability of a discarded censored trial is $1 - q_\lambda^4$. Its entire wasted dwell, including the successful waits before its first censor, has the exact expectation

$$\mathbb{E}[W_\lambda \mathbf{1}_{\text{censored}}] = \sum_{j=1}^{4} q_\lambda^{j-1} s \left[D + (j-1) \left(\frac{1}{\lambda} - \frac{Ds}{q_\lambda} \right) \right] \le 16 D e^{-\lambda D}. \tag{13}$$

The formulas follow by conditioning on the index of the first censored wait; $\mathbb{E}[T \mid T \le D] = \lambda^{-1} - Ds/q_\lambda$. Here $q_\lambda > 0$ because $D > 0$.

With the actual timestamp/deadline service, each normal trial has active dwell at most $4(D + \Delta)$ and expected active dwell at most $4/\lambda_- + 4\Delta$. A conservative addition $4\beta(D + \Delta)$ accounts for conditional bad-service branches, whose timeouts are also hard-capped. Discarded active dwell is bounded by $16(D + \Delta)e^{-\lambda_-(D-\Delta)} + 4\beta(D + \Delta)$ per trial. All pin, clamp, setup, settling and readout times are additional. These bounds include unsuccessful exposure; censoring does not make it free.

With $n = mq$ trials for the median version, or (11) trials for the direct mean, the worst-case active dwell is $4n(D + \Delta)$ and the expected active dwell is at most $n[4/\lambda_- + 4\Delta + 4\beta(D + \Delta)]$. The complete serial wall-time bound adds, for each trial,

$$C_{\text{coordinate}} + C_{\text{context}} + C_{\text{operating setting}} + 4(C_{\text{pin}} + C_{\text{settle}} + C_{\text{detector}} + C_{\text{timer}}) + C_{\text{arithmetic}},$$

plus common calibration, inter-menu finite slew and final deployment. Use the actual parallel service architecture to convert total work and component exposure to elapsed time; parallel copies are additional paid resources. No unpriced simultaneity is assumed. The existing safe mission preparation and deployment bill are separate from these local target-context assays and are not removed by a cheaper decision statistic.

The apparatus must detect the *first exit*, not merely read a late endpoint after possible returns. A failed event record may be flagged and charged in β; an unbounded missed-event mechanism cannot be silently reinterpreted as an exponential wait. If the requested exit floor is small, its numerical λ_-^{-1} exposure is retained.

2 A matched decision-exposure lower bound at the same local access

The quadratic accuracy cost cannot generally be removed by observing more of a local trajectory, choosing fields adaptively, or granting free parameter-independent pins. The following argument is an elementary continuous-time likelihood comparison, in the tradition of sequential change-of-measure lower bounds. It does not claim that inverse-gamma estimation or sequential testing is new.

Theorem 4 (Two-menu decision requires quadratic local exposure)**.** *Fix $0 < \eta \le 1/4$, $0 < \delta < 1/2$, $\lambda_0 > 0$, any $N \ge 1$, and a known $q_* \in (0, 1/4]$. There are two models of N independent bits and two operating menus such that:*

1. *both menus in both models have the same product invariant law with one-bit probability q_*, and the singleton target $b = 1^N$ has $p = q_*^N$;*
2. *every local* boundary recovery *rate $Q(b, b^{(i)})$ is in $[\lambda_0(1-2\eta), \lambda_0(1+2\eta)]$;*
3. *the two models reverse which menu minimizes the mean first-event time, and also reverse which maximizes it, with a multiplicative gap larger than $1+\eta$;*
4. *any adaptive decision procedure with error at most δ on both models must have expected total active local exposure*

$$\mathbb{E}T_{\text{exposure}} \ge \frac{(1-2\delta)\log((1-\delta)/\delta)}{64\lambda_0\eta^2} \tag{14}$$

under each model.

The procedure may observe complete continuous local jump paths, use arbitrary predictable calibrated fields, and select operating menus and starting bits adaptively. Free resets may reveal their commanded state but no additional parameter information. With parallel cells T_{exposure} is the sum of their active durations, not wall time. The permitted parameter-dependent observations are the local state/jump records; no extra rate-revealing actuator side-channel is supplied.

Proof. Put $\theta = 2\eta \le 1/2$ and $\lambda_\pm = \lambda_0(1 \pm \theta)$. In model zero, menu one has single-bit $1 \to 0$ recovery rate λ_+ and menu two has recovery rate λ_-. The corresponding $0 \to 1$ failure rate is $\lambda_\pm q_*/(1-q_*)$; for $q_* = 1/4$ it is $\lambda_\pm/3$. Swap the menus in model one. Both global generators are common scalar multiples of the same independent-bit generator. Their equilibrium-safe means therefore scale exactly inversely with their local flip rate. The ratio is $(1+\theta)/(1-\theta) > 1+\eta$ for every N, and the invariant target mass remains q_*^N. Given a strict reliable-family bound $\bar{q} < 1/2$, choosing $0 < q_* \le \min(\bar{q}/2, 1/4)$ places this hard pair strictly within that family.

To include arbitrary experimental fields, let a field z change the known conditional probability to $\sigma(\text{logit}(q_*) + z)$, while the attempt clock in that menu remains $\lambda_\pm/(1-q_*)$. This is a valid subclass of the existing access model. The two possible jump intensities at any history have the form $a = \lambda_0(1+\theta)v/(1-q_*)$ and $b = \lambda_0(1-\theta)v/(1-q_*)$ in one order, with the same known $v \in [0,1]$ determined by the field and current bit. For nonnegative intensities,

$$a\log(a/b) - a + b \le (a-b)^2/b \le \frac{4\lambda_0\theta^2}{(1-q_*)(1-\theta)} \le 16\lambda_0\theta^2 = 64\lambda_0\eta^2.$$

The same bound applies in the opposite direction; the zero-intensity limit is interpreted continuously. The controlled point-process likelihood chain rule integrates this divergence rate over the total active local exposure. Predictable choices and parameter-independent resets use the same conditional kernels under the two models and add no divergence. Consequently

$$\text{KL}(\mathsf{P}_0 \| \mathsf{P}_1) \le 64\lambda_0\eta^2\mathbb{E}_0 T_{\text{exposure}},$$

and conversely with the model indices exchanged. Bounded exposure gives this directly. For integrable adaptive stopping, stop first at a finite exposure cap and pass to the limit in the likelihood compensator.

A correct $(1+\eta)$ menu decision must select the appropriate menu in each model, so it is a binary test with both errors at most δ. Data processing to that decision gives

$$\text{KL}(\mathsf{P}_0 \| \mathsf{P}_1) \ge \text{kl}(1-\delta, \delta) = (1-2\delta)\log((1-\delta)/\delta).$$

Combine the inequalities in both directions. □

At fixed contrast, the positive assay's expected active exposure is $O(\lambda_-^{-1}\eta^{-2}\log(K/\delta))$, matching the lower bound's accuracy and confidence dependence for two menus up to constants and the stated contrast. The hard worst-case censor cap adds $\log(1/\eta)$; pins, service latencies and calibration are additional resources and are not lower-bounded by (14). The lower bound is uniform in the exponentially small global target probability. It does not assert optimal dependence on contrast, menu count, detector price or physical clock resolution.

3 Ordinary methods, the new interface and retained limits

The statistic $(n-1)/\sum_{j=1}^{n} T_j$ for an exponential rate is classical; the gamma integral above proves the needed $n = 4$ instance. Likelihood-based exponential-rate estimation is an appropriate strong ordinary comparator. A method that estimates every local rate can also reach quadratic precision cost per rate, but needlessly acquires N parameters when the decision requires only their average. Conversely, an ordinary method that groups repeated waits within a randomly chosen context and then averages the classical unbiased statistics implements the same direct representation and can match this result. No claim is made that every conventional estimator has cubic cost.

In particular, a small-time endpoint estimator that divides an observed exit probability by its hold time uses a time of order η to suppress linearization bias. Its reset/read count can scale as η^{-3} even though its total exposure already scales quadratically. The present grouped first-exit statistic removes that particular extra reset factor, retains a finite physical clock construction, and estimates the arithmetic rate average without a rate table. It does not improve on a stronger grouped-wait method already using this identity.

The new contribution for the chosen decision is the complete interface: requested operating contexts, an average of heterogeneous fast-side rates, finite censor/clip/timing errors including deadline boundary bands, dependence-safe confidence, all discarded exposure, and a matched controlled-observation lower bound. Its use in a global rare-event intervention still relies on the separate target-set and spectral or potential-theoretic decision theorem. Nothing in the local data certifies that unobserved failure mechanisms are absent.

4 Executed synthetic service and scaling record

The supplied program runs the finite direct-mean protocol at $\eta = 1/8$, $\lambda_- = 1/2$, $\lambda_+ = 6/5$, total $K = 6$ menu assays and $\delta = 0.05$. A rational upper count based on an integer $\log_2$ ceiling gives 49,152 trials per menu, at most 196,608 pin–release observations, $D = 22$, and $\Delta = 1.05964 \cdot 10^{-5}$ model-time units. The clipping threshold is $a = 5/12$, the output cap is $B = 36/5$, and all controls and arithmetic tolerances are recorded exactly as fractions.

The synthetic service includes a persistent detector latency $\Delta/4$, an actual censor deadline $D + \Delta/2$, a dyadic timestamp grid no larger than $\Delta/8$, output rounding, sequential early censor stops, and all discarded active dwell. Thus it executes the finite observations used by the proof; it is not a computation from an exposed kinetic table. Reciprocal quantization and final interval endpoints use exact integer and rational arithmetic; decimal interval displays are rounded outward. Hidden arrays define the simulator and are inspected only by the separate audit. Contexts are heterogeneous and indexed by a uniform coordinate request. The two menus have different actual operating clocks. The assumed copy fidelity, source model and bound calibration are not themselves acquired or tested by this synthetic execution.

N	Menu	True rate mean	Acquired mean	Paid active dwell
64	0	1.015459	1.011554	$1.96438 \cdot 10^5$
64	1	0.583779	0.583463	$3.40130 \cdot 10^5$
4096	0	0.998465	0.994689	$1.99658 \cdot 10^5$
4096	1	0.600447	0.600843	$3.30337 \cdot 10^5$
262144	0	0.999717	0.994509	$1.99813 \cdot 10^5$
262144	1	0.599923	0.598241	$3.31448 \cdot 10^5$

The relative confidence intervals are disjoint within every menu pair and select menu 1 for smaller mean escape rate. All observed relative errors are below 0.54%. This observed error is not a replacement for the 12.5% theorem guarantee. One censored trial in the largest menu-1 run, including its preceding successful waits, is present in the discarded-dwell ledger. No global first-event law or stationary rarity was computed in this local audit; the global decision bridge is a separate theorem.

These time units are defined by the supplied local rates. If the rates are interpreted in inverse seconds, an active dwell of $3.4 \cdot 10^5$ units is approximately four days on one serial channel, before pins, clamps, settling, detector/timer service, calibration, finite slew, safe mission preparation and deployment. No numerical price for those unprovided services is invented. The timing bound is sufficient, not a universal lower bound, and the statistical constants are conservative. The earlier κ^2 median-of-means schedule and its larger execution are retained as a valid development record; the final proof improves the conditional second-moment bound to $4\lambda_+\mu$ and supplies both the sharper median version and the direct-mean option.

High-precision integration checks the clipping loss and exact discarded dwell formulas at four rates. Separate algebra checks the likelihood upper bound and two-menu mean ratio at four accuracies. These are mathematical and synthetic service diagnostics, not a physical apparatus experiment.

Primary sources and attribution

The inverse-gamma unbiased exponential-rate identity is standard; an author-written derivation is Eduardo García-Portugués and Isabel Molina, *Statistical Inference*, Chapter 3, Example 3.4, `https://egarpor.github.io/SI-UC3M/point.html`. The proof here is included in full rather than imported as a new estimator. The confidence lower bound uses the same change-of-measure principle as *On the Complexity of Best-Arm Identification in Multi-Armed Bandit Models*, E. Kaufmann, O. Cappé, and A. Garivier, JMLR 17(1):1–42 (2016), `https://www.jmlr.org/papers/volume17/kaufman16a/kaufman16a.pdf`. The controlled continuous-time likelihood bound is derived explicitly for the present two-menu family; an iid bandit formula is not silently transported across physical access models. The direct-mean confidence argument is the classical martingale Bernstein/Freedman method, here proved with the needed constants; see D. A. Freedman, *On Tail Probabilities for Martingales*, Annals of Probability 3(1):100–118 (1975), DOI 10.1214/aop/1176996452.

The operating-context, faithful-copy, calibrated-field and priced-pin premises come from the unchanged acquired first-event package, especially its `experimental_operating.tex` source and the preserved active acquisition construction. This new assay uses their actual first-exit observation access, but does not execute the earlier uniformized operating-coin factory or infer a clock at an unmeasured operating field.